VENKATA SAI GANESH MANDA

(623)-955-9520 | saiganesh.profess@gmail.com | linkedin.com/in/sai-ganesh-manda-bo2002/ | github.com/mvsg2 | mvsg2.github.io

Target Roles: SDE | AI/ML Engineer | Robotics SDE | Data Scientist/Engineer (Fall 2026, Spring, Summer 2027)

SUMMARY

Master's student in Robotics and Autonomous Systems (AI) at Arizona State University (GPA 4.0) with hands-on experience building scalable distributed systems, LLM-powered multi-agent pipelines, cloud-native data pipelines, and robotics software integrating computer vision, motion planning, and autonomous control. Proven track record modifying open-source frameworks, writing automated test scripts, analyzing large datasets, and authoring technical documentation. Proficient in Python, C++, SQL, R, OOP, data structures and algorithms, CI/CD, and Agile SDLC. Passionate about owning the full software lifecycle from design through deployment across AI, robotics, and data engineering domains.

EDUCATION

M.S. Robotics and Autonomous Systems (AI), Arizona State University **Aug 2025 – May 2027**
SCAI, Ira A. Fulton Schools of Engineering, Tempe, AZ | GPA: 4.0/4.0
Relevant Coursework: Applied Linear Algebra for Engineers, Robotic Systems I, Agentic AI, Statistical Machine Learning

B.Tech. Artificial Intelligence and Data Science, IIITDM Kurnool **Sep 2020 – Jun 2024**
Exchange student, University of Agder (UiA), Grimstad, Norway | Jan 2024 – Jun 2024

TECHNICAL SKILLS AND CERTIFICATIONS

Languages: Python, C++, C, R, SQL, HTML, CSS, JavaScript (basic)
Software Eng: OOP, Data Structures and Algorithms, Distributed Systems, Design Patterns, Agile/Scrum, CI/CD, Git (CLI), Code Review, Debugging, Linux CLI, SDLC
AI/ML: PyTorch, TensorFlow, Scikit-learn, LangChain, vLLM, RAG, Multi-Agent Architectures, Anthropic/OpenAI/Gemini APIs, GenAI Tools, NLP
Robotics and CV: ROS 2, OpenCV, Computer Vision, Homography, Perspective Transforms, Affine Calibration, Motion Planning, Path Planning
Data Science: EDA, Statistical Modeling, Data Wrangling, Predictive Modeling, A/B Testing, Causal Inference, NumPy, Pandas, Hadoop, Spark
Testing and QA: Unit Testing, Smoke Testing, Pytest, Functional Testing, Test Automation, Defect Tracking, Validation and Verification
Cloud and Infra: AWS SageMaker, Azure AI Studio, REST APIs, SLURM, HPC, Microservices, Fault-Tolerant System Design
Certifications: ML Specialization (Coursera/Stanford); Big Data and Spark (EdX)

PROFESSIONAL EXPERIENCE

AI Intern | Semine AS, Kristiansand, Norway **Jan 2024 – Jun 2024**

- Designed and deployed a scalable, cloud-native document intelligence pipeline on Azure AI Studio integrating OCR, REST API-based field extraction, and automated accounting workflows, reducing manual processing time by 30% in production.
- Applied SQL queries and Pandas-based EDA over 300+ invoice records to benchmark pipeline performance, identify and debug data quality defects, and deliver data-driven reports to technical and business stakeholders.
- Fine-tuned document models across 200+ foreign-language invoice types using transfer learning, boosting extraction accuracy by 15% over the baseline OCR pipeline through iterative experimentation and structured validation.

PROJECTS

CodeLens: RAG-Based Codebase Intelligence API | Independent Project **May 2026 – Present**

- Shipped a RAG-based codebase intelligence API using hybrid BM25 and semantic retrieval with cross-encoder reranking, fixing a retrieval regression that raised hit rate from 85% to 100% on a hand-labeled 26-query evaluation set.
- Built a GitHub Actions CI pipeline gating merges on automated retrieval and LLM-judged quality checks, root-causing 3 distinct caching bugs that cut a 3.5-hour reindex step down to minutes on every run after the first.
- Achieved approximately 80% faithfulness and answer-relevancy scores on an automated LLM-judged quality gate, cutting judge token usage by approximately 33% and generation cost by approximately 96%, bringing full evaluation cost down to $0.62 per run for deployment gating. https://github.com/mvsg2/CodeLens

Tau-Bench: Multi-Agent LLM Infrastructure | Arizona State University **Jan 2026 – Present**

- Modified 11 source files in the open-source vLLM framework to enable distributed model serving on Intel Gaudi2 HPUs and NVIDIA A100 GPUs via Git CLI, eliminating external API dependency and reducing inference latency across 825 total runs (165 tasks x 5 trials).
- Authored smoke tests and unit tests using Pytest for critic agent evaluation logic in the LLM-Modulo framework; co-authored 115 SLURM scripts for fault-tolerant HPC execution across Qwen3 models from 4B to 32B parameters, with technical documentation for all workflows.
- Led structured defect analysis over 3,677 agent failures via systematic root-cause debugging and data analysis; co-designed a Planner-Critic-Executor architecture to resolve the dominant failure mode, improving system reliability across the full evaluation dataset. https://github.com/mvsg2/tau-bench-CSE598-repo

Autonomous Maze-Solving Robot | Arizona State University **Aug 2025 – Dec 2025**

- Built a modular perception-planning-control pipeline in Python using OpenCV (Otsu thresholding, morphological filtering, contour detection, perspective transform homography) to convert raw camera frames into structured occupancy grids, following OOP design patterns with independently testable components managed via Git CLI.
- Integrated Gemini 2.5 Flash via agentic LLM tool calls for automated entrance/exit detection, then applied optimization-based path planning (BFS and A* with Manhattan distance heuristic) to compute minimum-length waypoint sequences for autonomous robot execution.
- Calibrated pixel-to-robot coordinate mapping via a 6-parameter affine least-squares fit, achieving a 100% success rate across all tested 4x4 maze configurations regardless of camera orientation. https://github.com/mvsg2/RAS-545-Midterm-Project

LLM-Controlled Robotic Manipulation | Arizona State University **Aug 2025 – Dec 2025**

- Architected a modular agentic pipeline using Anthropic Claude REST API and ROS 2, integrating OpenCV for scene perception and object detection to enable a Dobot Magician Lite to autonomously execute pick-and-place tasks from natural language commands.
- Implemented a Python tool-calling interface computing affine transforms and homographies for pixel-to-world coordinate conversion, with clean OOP separation of concerns across vision, planning, and execution modules and full codebase managed via Git CLI.
- Validated end-to-end autonomous manipulation via operational integration testing across varied object placements, achieving zero manual intervention across all evaluated scenarios; published open-source with documentation. https://github.com/mvsg2/RAS-545-Final-Project